

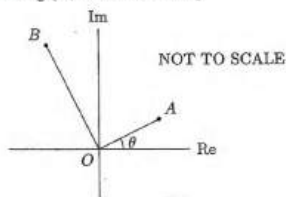
HSC Extension 2 Mathematics

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Complex Number



The points A and B in the diagram represent the complex numbers z_1 and z_2 respectively, where $|z_1| = 1$ and $\arg(z_1) = \theta$ and $z_2 = \sqrt{3}iz_1$



Which of the following represents $z_2 - z_1$?

- A) $2e^{i(\frac{2\pi}{3} + \theta)}$ B) $3e^{i(\frac{2\pi}{3} + \theta)}$ C) $2e^{i(\frac{2\pi}{3} - \theta)}$ D) $3e^{i(\frac{2\pi}{3} - \theta)}$



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Complex Number



Express $-2 + 2\sqrt{3}i$ in the form $re^{i\theta}$

2



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Complex Number



- (i) Use Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ to find a similar simplified result for $e^{-i\theta}$ 1
- (ii) Hence show $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$ 1
- (iii) Use the result from part (ii) to show that $\sin^3\theta = \frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta$ 3



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Complex Number



The polynomial $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$ has zeros $a + bi$ and $a + 2bi$ 3
where a and $b \in \mathbb{R}$ and $b \neq 0$.

By evaluating a and b , find all the zeros of $P(x)$.



Complex Number Solution

$$\begin{aligned} -2 + 2\sqrt{3}i & \quad \arg = \tan^{-1} \frac{2\sqrt{3}}{-2} \\ r = \sqrt{(-2)^2 + (2\sqrt{3})^2} & = \tan^{-1}(-\sqrt{3}) \\ = \sqrt{4+12} & \quad \text{Working angle is } \frac{\pi}{3} \\ = 4 \text{ ①} & \quad \text{2nd quadrant } \therefore \frac{2\pi}{3} \text{ ①} \\ \text{so } -2 + 2\sqrt{3}i & = 4e^{\frac{2\pi i}{3}} = 4e^{i(\frac{2\pi}{3})} \text{ ②} \end{aligned}$$



Complex Number Solution

$$\begin{aligned} z_2 - z_1 & \\ = \sqrt{3}iz_1 - z_1 & \\ = \text{cis } \theta (-1 + \sqrt{3}i) & \\ = \text{cis } \theta \times 2 \text{cis}(-\frac{\pi}{3}) & \\ = 2 \text{cis}(\theta - \frac{2\pi}{3}) & \\ = 2e^{i(\theta - \frac{\pi}{3})} & \\ A & \end{aligned}$$



Complex Number Solution

$$\begin{aligned} 1) P(x) &= x^4 - 4x^3 + 11x^2 - 14x + 10 = (x^2 - 2ax + (a^2 + b^2))x^2 \\ & \quad (x^2 - 2ax + (a^2 + 4b^2)) \text{ ①} \\ \text{Equating coefficients of } x^3 & \\ -4ax^3 &= -4ax^3 \therefore a=1 \text{ ②} \\ \text{Equate constants} & \\ 10 &= (a^2 + b^2)(a^2 + 4b^2) \\ 10 &= (b^2 + 1)(4b^2 + 1) \quad \begin{array}{r} +15 \\ \times 36 \\ \hline 9-4 \end{array} \\ 0 &= 4b^4 + 5b^2 - 9 \\ 0 &= 4b^4 + 9b^2 - 4b^2 - 9 \\ 0 &= b^2(4b^2 + 9) - (4b^2 + 9) \\ 0 &= (b^2 - 1)(4b^2 + 9) \\ \therefore b &= \pm 1 \end{aligned}$$

$\therefore$ Roots/zeros are $1 \pm i, 1 \pm 2i$ ③



Complex Number Solution

$$\begin{aligned} \text{ii) } e^{i\theta} &= \cos \theta + i \sin \theta \\ e^{-i\theta} &= \cos(-\theta) + i \sin(-\theta) \\ e^{-i\theta} &= \cos \theta - i \sin \theta \\ \text{iii) } e^{i\theta} - e^{-i\theta} &= (\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta) \\ &= 2i \sin \theta \\ \therefore \sin \theta &= \frac{1}{2i}(e^{i\theta} - e^{-i\theta}) \\ \text{iv) } \sin \theta &= \frac{1}{2i}(e^{i\theta} - e^{-i\theta}) \\ \sin^2 \theta &= \frac{1}{8i^2}(e^{i\theta} - e^{-i\theta})^2 \text{ ①} \\ &= \frac{1}{8i^2}(e^{3i\theta} - 3e^{i\theta}e^{-i\theta} + 3e^{i\theta}e^{-2i\theta} - e^{-3i\theta}) \\ &= \frac{1}{8i^2}(e^{3i\theta} - e^{-3i\theta} - 3(e^{i\theta} - e^{-i\theta})) \text{ ②} \\ &= \frac{1}{8i^2}(2i \sin 3\theta - 3 \times 2i \sin \theta) \\ &= \frac{3}{4} \sin \theta - \frac{1}{4} \sin 3\theta \text{ ③ as required.} \end{aligned}$$



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Complex Number

Given two complex numbers, $z = -3 + 4i$ and $w = 1 - 3i$, find:

(i) $z - 2w$

1

(ii) zw

1

(iii) $\frac{1}{z}$

1

(iv) $z + \overline{iw}$

1



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Complex Number

Use de Moivre's Theorem to simplify $\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^5 - \left(\frac{\sqrt{3}}{2} - \frac{i}{2}\right)^5$

2



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Complex Number

The polynomial $P(z) = z^4 - 6z^3 + 14z^2 - 24z + 40$ has four complex zeroes, two of which are $2i$ and $3 - i$.

(i) Write down the other two zeroes, giving a clear justification for your answer.

1

(ii) Hence write $P(z)$ as a product of two quadratic factors with real coefficients.

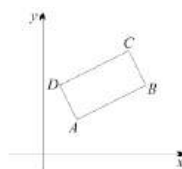
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Complex Number



The diagram above shows a rectangle $ABCD$ in the Argand diagram. The rectangle is twice as long as it is wide. The points A and B represent the complex numbers $2 + 2i$ and $6 + 4i$ respectively. Find the complex numbers corresponding to each of the following:

(i) $\overline{AB}$

1

(ii) $\overrightarrow{AD}$

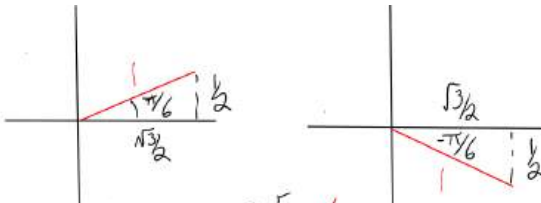
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(iii) point C

1



Complex Number Solution



$$\begin{aligned}
 & \left(\operatorname{cis} \frac{\pi}{6} \right)^5 - \left(\operatorname{cis} \left(-\frac{\pi}{6} \right) \right)^5 \quad \checkmark \\
 &= \operatorname{cis} \left(\frac{5\pi}{6} \right) - \operatorname{cis} \left(-\frac{5\pi}{6} \right) \\
 &= \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) - \left(\cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6} \right) \\
 &= 2i \sin \frac{5\pi}{6} \quad \checkmark \quad \left(\text{since } \sin \frac{5\pi}{6} = \sin \frac{\pi}{6} = \frac{1}{2} \right) \\
 &= i
 \end{aligned}$$



Complex Number Solution

$$\begin{aligned}
 \text{(i)} \quad & (-3+4i) - 2(1-3i) = -5+10i \quad \checkmark \\
 \text{(ii)} \quad & (-3+4i)(1-3i) = -3+4i+9i-12i^2 \\
 &= 9+13i \quad \checkmark \\
 \text{(iii)} \quad & \frac{1}{-3+4i} \times \frac{-3-4i}{-3-4i} = \frac{1}{25}(-3-4i) \quad \checkmark \\
 \text{(iv)} \quad & \overline{z+iu} = \overline{-3+4i+i(1-3i)} \\
 &= \overline{0+5i} \\
 &= -5i \quad \checkmark
 \end{aligned}$$



Complex Number Solution

$$\begin{aligned}
 \text{(i)} \quad & \vec{OA} \text{ is } 2+2i \quad \vec{AB} = \vec{OB} - \vec{OA} \quad \checkmark \\
 & \vec{OB} \text{ is } 6+4i \quad \vec{AB} \text{ is } 4+2i \quad \checkmark \\
 \text{(ii)} \quad & \vec{AD} = \vec{BC} \quad (\text{Equal Parallel sides of rectangle}) \\
 & \vec{AD} \text{ is } \vec{BC} \text{ rotated } 90^\circ \text{ anticlockwise and halved.} \\
 & \vec{AB} \text{ is } \frac{i}{2} \times (4+2i) = -1+2i \quad \checkmark \\
 \text{(iii)} \quad & \vec{OC} = \vec{OB} + \vec{BC} \quad C \text{ is } 6+4i + (-1+2i) \quad \checkmark \\
 &= \vec{OB} + \vec{AB} \quad = 5+6i \quad \checkmark
 \end{aligned}$$



Complex Number Solution

$$\begin{aligned}
 \text{(i)} \quad & \text{Other two roots are } -2i \text{ and } 3+i \\
 & \text{Since complex roots appear in conjugate pairs when coefficients of polynomial are REAL} \quad \checkmark \\
 & \text{Reason and Answer} \\
 \text{(ii)} \quad & P(z) = (z+2i)(z-2i)(z-(3-i))(z-(3+i)) \quad \checkmark \\
 &= (z^2+4)(z^2-6z+10) \\
 & \quad \left(\text{Check: } z^4 - 6z^3 + 10z^2 + 4z^2 - 24z + 40 \right. \\
 & \quad \quad \left. = z^4 - 6z^3 + 14z^2 - 24z + 40 \right)
 \end{aligned}$$



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Complex Number



Consider the following pair of inequations in the complex plane.

$$|z - 2 - i| \leq 1 \quad 0 \leq \text{Arg } z < \frac{\pi}{4}$$

- (i) Sketch the intersection of the regions given by the two inequations, making sure to clearly indicate whether the boundaries and their points of intersection are included in the required region. [3]
- (ii) Hence calculate the maximum value of $|z|$. [1]



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Complex Number



- (i) Use proof by induction to prove de Moivre's Theorem, that [3]

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

for all integers $n \geq 1$.

- (ii) Hence show that $\frac{(\cos \frac{\pi}{18} + i \sin \frac{\pi}{18})^{11}}{(\cos \frac{\pi}{36} + i \sin \frac{\pi}{36})^4}$ is purely imaginary. [1]



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Complex Number



Consider $\frac{(3 + 4i)(1 + 2i)}{1 + 3i} = (1 + i)q$, where q is a real number.

- (i) Find the value of q . [1]
- (ii) Hence explain why $\tan^{-1} \left(\frac{4}{3} \right) + \tan^{-1} 2 - \tan^{-1} 3 = \frac{\pi}{4}$. [1]



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Complex Number



- (i) By applying de Moivre's theorem to $(\cos \theta + i \sin \theta)^3$, show that [1]

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta.$$

- (ii) Hence, or otherwise, find an expression for $\tan 3\theta$ in terms of $\tan \theta$. [2]

- (iii) Show that $\tan \frac{\pi}{12}$ is a root of the cubic equation [2]

$$x^3 - 3x^2 - 3x + 1 = 0$$

and find two other values of θ for which $\tan \theta$ is a root of this equation, where $0 < \theta < \pi$.

- (iv) Hence show that [1]

$$\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} = 4.$$



Complex Number Solution

(i) Show conjecture true for $n=1$

$$\text{LHS} = (\cos \theta + i \sin \theta)^1 = \cos \theta + i \sin \theta$$

$$\text{RHS} = \cos \theta + i \sin \theta$$

$$\text{LHS} = \text{RHS} \quad \text{hence true for } n=1$$

Assume true for $n=k$, $k \in \mathbb{Z}^+$ that is,

$$(\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$$

RTP for $n=k+1$

$$(\cos \theta + i \sin \theta)^{k+1} = \cos(k+1)\theta + i \sin(k+1)\theta \quad \text{(*)}$$

$$\text{LHS} = (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta)$$

$$= (\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta) \quad \text{(by assumption (*))}$$

$$= \cos k\theta \cos \theta - \sin k\theta \sin \theta + i(\sin k\theta \cos \theta + \cos k\theta \sin \theta)$$

$$= \cos(k+1)\theta + i \sin(k+1)\theta \quad \text{(by compound angle theorems)}$$

$$= \text{RHS}$$

hence if true for $n=k$ then true for $n=k+1$ By principle of mathematical induction conjecture true for all positive integer n .

$$(ii) \frac{(\cos \frac{\pi}{18})^4}{(\cos \frac{\pi}{36})^4} = \frac{\cos(\frac{4\pi}{18})}{\cos(\frac{\pi}{9})}$$

$$= \frac{\cos(\frac{4\pi}{18} - \frac{2\pi}{18})}{\cos(\frac{\pi}{9})}$$

$$= \frac{\cos \frac{\pi}{2}}{\cos(\frac{\pi}{9})}$$

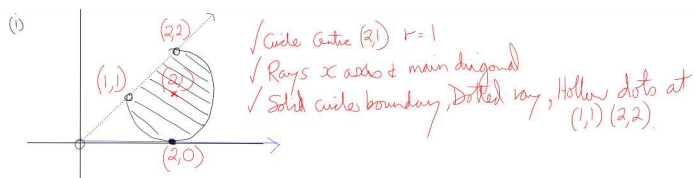
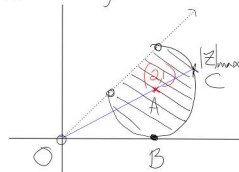
$$= \frac{0}{\cos(\frac{\pi}{9})}$$

$$= 0$$

$$\therefore \text{purely imaginary}$$



Complex Number Solution

(ii) Max value of $|z|$ occurs for ray from origin passing through centre of circle i.e. pt C

$$OA = \sqrt{1^2 + 1^2}$$

$$= \sqrt{2}$$

$$OC = 1 + \sqrt{2}$$

$$|z|_{\max} = 1 + \sqrt{2} \quad \checkmark$$



Complex Number Solution

$$(i) (\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3\cos^2 \theta i \sin \theta + 3\cos \theta i^2 \sin^2 \theta + i^3 \sin^3 \theta$$

$$\cos 3\theta + i \sin 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta + i(3\cos^2 \theta \sin \theta - \sin^3 \theta) \quad \text{✓ show}$$

$$\text{Equating real parts } \cos 3\theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$$

$$(ii) \text{Equating imag parts } \sin 3\theta = 3\cos^2 \theta \sin \theta - \sin^3 \theta$$

$$\text{hence } \tan 3\theta = \frac{\sin 3\theta}{\cos 3\theta}$$

$$= \frac{3\cos^2 \theta \sin \theta - \sin^3 \theta}{\cos^3 \theta - 3\cos \theta \sin^2 \theta} \quad \text{(*)}$$

$$= \frac{3\tan \theta - \tan^3 \theta}{1 - 3\tan^2 \theta} \quad \checkmark$$

$$(iii) \tan 3\theta = 1$$

$$3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12} = \frac{3\pi}{4}$$

$$\text{Let } x = \tan \frac{\pi}{12}$$

$$1 = \frac{3x - x^3}{1 - 3x^2}$$

$$1 - 3x^2 = 3x - x^3$$

$$x^3 - 3x^2 - 3x + 1 = 0 \quad \text{as reqd} \quad \checkmark \text{ show}$$

$$\text{Roots of cubic equation } x = \tan \theta \quad \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{3\pi}{4} \quad \checkmark$$

$$(iv) \sum x = -\frac{b}{a} \quad \text{hence } \tan \frac{\pi}{12} + \tan \frac{5\pi}{12} + \tan \frac{3\pi}{4} = -\frac{(-3)}{1} \quad \checkmark \text{ show}$$

$$\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} - 1 = 3$$

$$\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} = 4 \quad \text{(as required)}$$



Complex Number Solution

$$(i) q = \frac{(3+4i)(1+2i)}{(1+3i)(1+i)}$$

$$= \frac{3-8+4i+6i}{1-3+3i+i}$$

$$= \frac{10i-5}{4i-2}$$

$$= \frac{5(-1+2i)}{2(-1+2i)}$$

$$q = \frac{5}{2} \quad \checkmark$$

$$(ii) \text{From (i) } \text{Arg} \left[\frac{(3+4i)(1+2i)}{(1+3i)} \right] = \text{Arg} \left[\frac{5}{2}(1+i) \right]$$

$$\text{Arg}(3+4i) + \text{Arg}(1+2i) - \text{Arg}(1+3i) = \frac{\pi}{4} \quad \checkmark \text{ show}$$

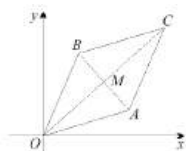
$$\tan^{-1}\left(\frac{4}{3}\right) + \tan^{-1}\left(\frac{2}{1}\right) - \tan^{-1}\left(\frac{3}{1}\right) = \frac{\pi}{4}$$



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Complex Number



Let $z_1 = \text{cis } \alpha$ and $z_2 = \text{cis } \beta$ where $0 < \alpha < \beta < \frac{\pi}{2}$. The complex numbers z_1 , z_2 and $(z_1 + z_2)$ are represented by the points A , B and C respectively in the Argand diagram above. Let $(z_1 + z_2) = r \text{cis } \theta$ and let OC and AB intersect at M .

- State what type of quadrilateral $OACB$ is, giving a clear geometric justification for your answer. 1
- Use the properties of the quadrilateral to determine θ in terms of α and β . 1
- By considering triangle OAC , express r in terms of α and β . 2
- Hence show that 2

$$\cos \alpha + \cos \beta = 2 \cos \left(\frac{\beta - \alpha}{2} \right) \cos \left(\frac{\beta + \alpha}{2} \right).$$



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Complex Number



Sketch the curve described by $\frac{1}{z} + \frac{1}{\bar{z}} = 1$ on the complex plane. 3



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Complex Number



Find the complex roots of $z^3 + 8i = 0$, and show them on an Argand diagram. 3



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Complex Number



Evaluate: 2

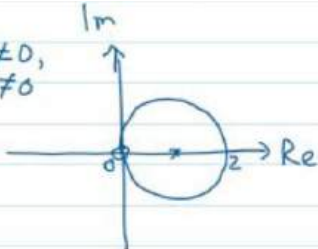
$$1 + e^{\frac{2\pi i}{3}} + \left(e^{\frac{2\pi i}{3}} \right)^2 + \left(e^{\frac{2\pi i}{3}} \right)^3 + \dots + \left(e^{\frac{2\pi i}{3}} \right)^{3n}.$$



Complex Number Solution

$$\frac{1}{z} + \frac{1}{\bar{z}} = 1, \quad z \neq 0, \quad \bar{z} \neq 0$$

$$\frac{z + \bar{z}}{z\bar{z}} = 1$$

$$\frac{2x}{x^2 + y^2} = 1$$


$$2x = x^2 + y^2$$

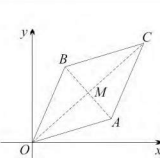
$$x^2 - 2x + y^2 = 0$$

$$(x-1)^2 - 1 + y^2 = 0$$

$$(x-1)^2 + y^2 = 1$$



Complex Number Solution



(i) OACB is a RHOMBUS since a parallelogram with equal adjacent sides $OA = OB = 1$ since $z + \bar{z}$ have unit moduli ✓

(ii) Diagonals of a rhombus bisect angles at vertices
hence $\theta = \alpha + \frac{1}{2}(\beta - \alpha)$
 $= \frac{1}{2}(\alpha + \beta)$ ✓

(iii) Cosine rule in $\triangle AOC$
 $OC^2 = OA^2 + AC^2 - 2 \times OA \times AC \times \cos \angle OAC$
 $= 1^2 + 1^2 - 2 \times 1 \times 1 \times [-\cos(\beta - \alpha)]$ ✓
 $OC^2 = 2 + 2\cos(\beta - \alpha)$ (since $AC = OB = 1$)
 $\angle OAC = 180^\circ - \angle BOA$ (Co-interior $\angle$, $OB \parallel AC$)
 $= 180^\circ - (\beta - \alpha)$
 $\cos \angle OAC = -\cos(\beta - \alpha)$
 $OC = 2\cos\left(\frac{\beta - \alpha}{2}\right)$ (Using Double Angle $\cos 2\theta = 2\cos^2\theta - 1$)
 $r = 2\cos\left(\frac{\beta - \alpha}{2}\right)$ (since supplementary angles opposite cosines)

(iv) since $z_1 + z_2 = \cos \alpha + i \sin \alpha + \cos \beta + i \sin \beta = 2\cos\left(\frac{\beta - \alpha}{2}\right) \left[\cos \theta + i \sin \theta \right]$ ✓
 $\cos \alpha + i \sin \alpha + \cos \beta + i \sin \beta = 2\cos\left(\frac{\beta - \alpha}{2}\right) \left[\cos \theta + i \sin \theta \right]$ ✓ show
 Equating real parts
 $\cos \alpha + \cos \beta = 2\cos\left(\frac{\beta - \alpha}{2}\right) \cos\left(\frac{\alpha + \beta}{2}\right)$



Complex Number Solution

Let $w = e^{\frac{2\pi i}{3}}$

$$1 + w + w^2 + \dots + w^{3n}$$


$$1 + w + w^2 = 0$$

$\therefore$ each sum of 3 consecutive terms is zero.

$$\therefore S = (1 + w + w^2) + w^3(1 + w + w^2) + \dots + w^{3n}$$

$$= w^{3n}$$

$$= 1$$



Complex Number Solution

$$z^3 + 8i = 0$$

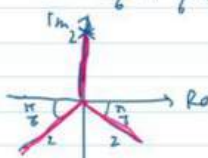
$$z^3 = -8i$$

$$r^3 e^{3i\theta} = 2^3 e^{-i\frac{\pi}{2}}$$

$$r = 2, \quad 3\theta = -\frac{\pi}{2} + 2n\pi$$

$$\theta = \frac{-\frac{\pi}{2} + 2n\pi}{3}$$

$$\theta = \frac{(4n-1)\pi}{6} \quad -\pi < \theta \leq \pi$$

$$\theta = -\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}$$


i.e. $z = 2i, 2e^{-\frac{\pi i}{6}}, 2e^{-\frac{5\pi i}{6}}$



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Complex Number



- (i) Prove that $e^{in\theta} + e^{-in\theta} = 2\cos n\theta$. 1
- (ii) Use part (i) and the binomial expansion of $(e^{i\theta} + e^{-i\theta})^3$ to prove that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$. 2
- (iii) Hence find all solutions to $8x^3 - 6x - 1 = 0$. 3



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Complex Number



Let P and Q be points representing the complex numbers z_1 and $z_2 = -2 + 6i$ respectively.

- (i) If $\triangle POQ$ forms an equilateral triangle with $\arg z_2 > \arg z_1$, find the complex number represented by P , in the form $a + ib$. 3
- (ii) Hence find the area of the $\triangle POQ$. 1



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Complex Number



Sketch the region on the Argand diagram defined by $|z - 3i| > |z + 2|$. 3



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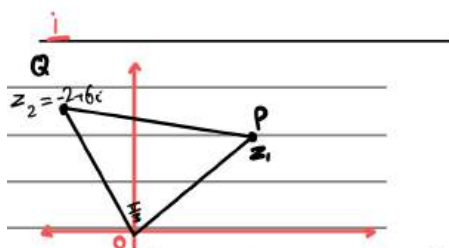
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Complex Number



Find α and β given that $z^3 + 6z - 4\sqrt{2}i = (z - \alpha)^2(z - \beta)$. 3





$$\begin{aligned} z_1 &= z_2 \times \left(\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \right) \\ &= (-2 + 6i) \times \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) \\ &= -1 + 3i + \sqrt{3}i + 3\sqrt{3} \\ &= 3\sqrt{3} - 1 + i(3 + \sqrt{3}) \end{aligned}$$

$$\begin{aligned} \text{ii) } |z_1| &= |z_2| = |-2 + 6i| \\ &= \sqrt{4 + 36} \\ &= \sqrt{40} \\ &= 2\sqrt{10} \end{aligned}$$

$$\begin{aligned} \therefore \text{Area} &= \frac{1}{2} \times 2\sqrt{10} \times 2\sqrt{10} \times \sin\frac{\pi}{3} \\ &= 20 \times \frac{\sqrt{3}}{2} \\ &= 10\sqrt{3} \text{ units}^2 \end{aligned}$$



$$\begin{aligned} \text{i) } e^{in\theta} + e^{-in\theta} &= \cos(n\theta) + i\sin(n\theta) + \cos(n\theta) - i\sin(n\theta) \\ &= 2\cos(n\theta) \end{aligned}$$

$$\text{OR } e^{in\theta} + e^{-in\theta} = 2\operatorname{Re}(e^{in\theta})$$

$$\begin{aligned} \text{ii) } (e^{i\theta} + e^{-i\theta})^3 &= e^{3i\theta} + 3e^{i\theta} + 3e^{-i\theta} + e^{-3i\theta} \\ &= (e^{3i\theta} + e^{-3i\theta}) + 3(e^{i\theta} + e^{-i\theta}) \\ (2\cos\theta)^3 &= 2\cos 3\theta + 6\cos\theta \end{aligned}$$

$$\begin{aligned} 8\cos^3\theta &= 2\cos 3\theta + 6\cos\theta \\ \cos 3\theta &= 4\cos^3\theta - 3\cos\theta \end{aligned}$$

$$\begin{aligned} \text{iii) } 8x^3 - 6x - 1 &= 0 \\ 2(4x^3 - 3x) &= 1 \end{aligned}$$

$$\text{Let } x = \cos\theta$$

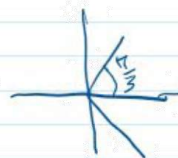
$$2(4\cos^3\theta - 3\cos\theta) = 1$$

$$2(\cos 3\theta) = 1$$

$$\cos 3\theta = \frac{1}{2}$$

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$



$$\therefore x = \cos\frac{\pi}{9}, \cos\frac{5\pi}{9}, \cos\frac{7\pi}{9} \text{ are solutions.}$$

(Note that the eqn must have three



For

$$z^3 + 6z - 4\sqrt{2}i = (z - \alpha)^2(z - \beta) \quad \text{--- (1)}$$

RHS of (1)

$$\begin{aligned} &(z^2 - 2\alpha z + \alpha^2)(z - \beta) \\ &= z^3 - 2\alpha z^2 + \alpha^2 z - \beta z^2 + 2\alpha\beta z - \alpha^2\beta \\ &= z^3 + z^2(-2\alpha - \beta) + z(\alpha^2 + 2\alpha\beta) - \alpha^2\beta \end{aligned}$$

From (1)

$$z^3 + 6z - 4\sqrt{2}i = z^3 + z^2(-2\alpha - \beta) + z(\alpha^2 + 2\alpha\beta) - \alpha^2\beta$$

Equating coefficients:

$$z^2: -2\alpha - \beta = 0$$

$$\beta = -2\alpha \quad \text{--- (2)}$$

$$z: \alpha^2 + 2\alpha\beta = 6 \quad \text{--- (3)}$$

$$\text{constant: } \alpha^2\beta = 4\sqrt{2}i \quad \text{--- (4)}$$

Now sub (2) into (3)

$$\alpha^2 + 2\alpha(-2\alpha) = 6$$

$$\alpha^2 - 4\alpha^2 = 6$$

$$-3\alpha^2 = 6$$

$$\alpha^2 = -2$$

$$\alpha = \pm\sqrt{2}i$$

$$\text{Let } P(z) = z^3 + 6z - 4\sqrt{2}i$$

Testing both values of α

$$\text{when } \alpha = \sqrt{2}i, P(\sqrt{2}i) = (\sqrt{2}i)^3 + 6(\sqrt{2}i) - 4\sqrt{2}i = -2\sqrt{2}i + 2\sqrt{2}i = 0$$

$$\alpha = -\sqrt{2}i, P(-\sqrt{2}i) = 2\sqrt{2}i - 6\sqrt{2}i - 4\sqrt{2}i \neq 0$$

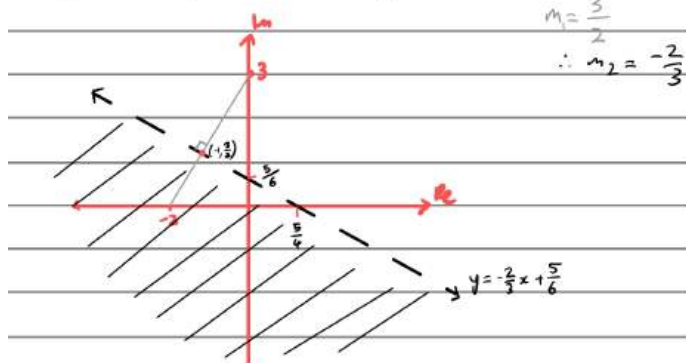
$\therefore \alpha = \sqrt{2}i$ is the double root

$$\text{Sub in (2) } \beta = -2\sqrt{2}i$$



$$|z - 3i| > |z - -2|$$

$$\text{i.e. } \left[\begin{array}{l} \text{the distance} \\ \text{from } 3i \end{array} \right] > \left[\begin{array}{l} \text{the distance} \\ \text{from } -2 \end{array} \right]$$



$$\begin{aligned} m_1 &= \frac{3}{2} \\ \therefore m_2 &= -\frac{2}{3} \end{aligned}$$



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Complex Number



- (i) Show that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$, where $\cos \theta \neq 0, n \in \mathbb{Z}^+$. 2
- (ii) By using the result in part (i), and by letting $z = i \tan \theta$, show that the roots of $(1 + z)^2 + (1 - z)^2 = 0$ are $z = \pm i \tan \frac{\pi}{4}$. 2



HSC Extension 2 Mathematics

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Complex Number



Let the complex number $\alpha = \frac{1}{2}e^{i\theta}$, where θ is real.

- (i) Show that the real part of the series 3

$$1 + \frac{1}{2}\alpha^3 + \frac{1}{4}\alpha^6 + \frac{1}{8}\alpha^9 + \dots \text{ is } \frac{256 - 16 \cos 3\theta}{257 - 32 \cos 3\theta}.$$

- (ii) Find an expression for 2

$$\frac{1}{2^4} \sin 3\theta + \frac{1}{2^8} \sin 6\theta + \frac{1}{2^{12}} \sin 9\theta + \dots \text{ in terms of } \sin 3\theta \text{ and } \cos 3\theta.$$



HSC Extension 2 Mathematics

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Complex Number



- (i) Show that $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$, where $\cos \theta \neq 0, n \in \mathbb{Z}^+$. 2
- (ii) Hence, show that the roots of $(1 + z)^2 + (1 - z)^2 = 0$ are $z = \pm i \tan \frac{\pi}{4}$ when z is purely imaginary. 2



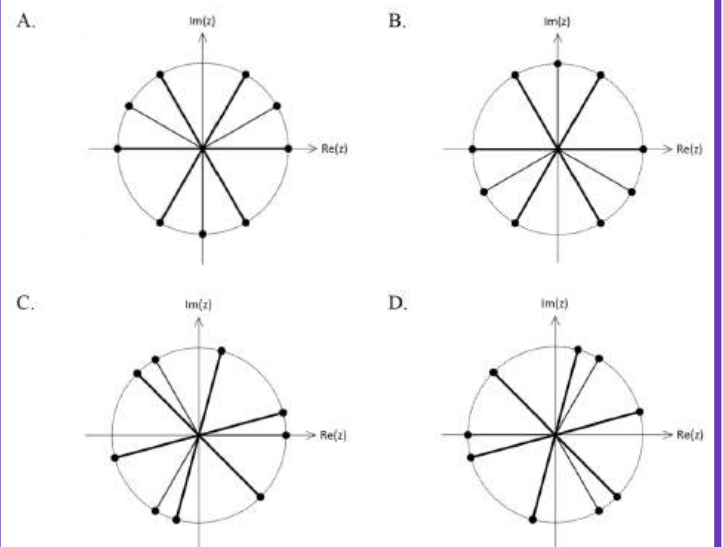
HSC Extension 2 Mathematics

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Complex Number



Which Argand diagram shows the solutions of $(z^3 - i)(z^6 - 1) = 0$?



Complex Number Solution

$$\begin{aligned} \frac{T_2}{T_1} &= \frac{\frac{1}{2}d^3}{1} = \frac{T_3}{\frac{1}{2}d^3} \\ \frac{1}{2}d^3 &= \frac{1}{2}d^3 \\ \therefore \text{AP with } r &= \frac{1}{2}d^3 \\ &= \frac{1}{2} \left(\frac{1}{2}e^{i\theta} \right)^3 \\ &= \frac{1}{16}e^{i3\theta} \\ \therefore S_{\infty} &= \frac{a}{1-r} = \frac{16}{1 - \frac{1}{16}e^{i3\theta}} = \frac{16}{1 - (\cos 3\theta + i\sin 3\theta)} \\ &= \frac{16(1 - \cos 3\theta) - i\sin 3\theta}{16[(1 - \cos 3\theta) + i\sin 3\theta]} \\ &= \frac{[(16 - \cos 3\theta) - i\sin 3\theta][(16 - \cos 3\theta) + i\sin 3\theta]}{16[(1 - \cos 3\theta)^2 - i^2 \sin^2 3\theta]} \\ &= \frac{256 - 16\cos 3\theta + 16i\sin 3\theta + \cos^2 3\theta + \sin^2 3\theta}{16(1 - \cos 3\theta)^2 + 16\sin^2 3\theta} \\ &= \frac{256 - 16\cos 3\theta + 16i\sin 3\theta + \cos^2 3\theta + \sin^2 3\theta}{16(1 - \cos 3\theta)^2 + 16\sin^2 3\theta} \\ &= \frac{256 - 16\cos 3\theta + 16i\sin 3\theta + 1}{16(1 - \cos 3\theta)^2 + 16\sin^2 3\theta} \\ &= \frac{257 - 16\cos 3\theta + 16i\sin 3\theta}{16(1 - \cos 3\theta)^2 + 16\sin^2 3\theta} \\ \therefore \text{real part is } &\frac{257 - 16\cos 3\theta}{16(1 - \cos 3\theta)^2 + 16\sin^2 3\theta} \\ \text{ii } T_2 &= \frac{1}{16}e^{i2\theta} = \frac{1}{24}(\cos 3\theta + i\sin 3\theta) = \frac{1}{24}\cos 3\theta + \frac{i}{24}\sin 3\theta \\ T_3 &= \frac{1}{256}e^{i6\theta} = \frac{1}{256}(\cos 6\theta + i\sin 6\theta) = \frac{1}{256}\cos 6\theta + \frac{i}{256}\sin 6\theta \\ \therefore \frac{1}{24}\sin 3\theta + \frac{i}{256}\sin 6\theta + \dots &\text{ is the imaginary part of the series} \\ \therefore \frac{1}{24}\sin 3\theta + \frac{i}{256}\sin 6\theta + \frac{i}{256}\sin 9\theta + \dots &= \frac{16\sin 3\theta}{257 - 16\cos 3\theta} \quad (\text{from part i}) \end{aligned}$$



Complex Number Solution

$$\begin{aligned} \text{i) LHS} &= (1 + i\tan\theta)^n + (1 - i\tan\theta)^n \\ &= \left(\frac{\cos\theta + i\sin\theta}{\cos\theta} \right)^n + \left(\frac{\cos\theta - i\sin\theta}{\cos\theta} \right)^n \\ &= \frac{(\cos\theta + i\sin\theta)^n}{\cos^n\theta} + \frac{(\cos\theta - i\sin\theta)^n}{\cos^n\theta} \\ &= \frac{\cos^n\theta + i\sin^n\theta + \cos^n\theta - i\sin^n\theta}{\cos^n\theta} \\ &= \frac{2\cos^n\theta}{\cos^n\theta} = \text{RHS} \\ \text{ii) Let } z &= i\tan\theta \quad \text{--- (1)} \\ \text{Now given that} \\ 0 &= (1+z)^2 + (1-z)^2 \\ 0 &= (1+i\tan\theta)^2 + (1-i\tan\theta)^2 \\ 0 &= \frac{2\cos 2\theta}{\cos^2\theta} \quad \text{from (i)} \\ \therefore 2\cos 2\theta &= 0 \\ \cos 2\theta &= 0 \\ 2\theta &= \frac{\pi}{2}, \frac{3\pi}{2}, \dots \\ \theta &= \frac{\pi}{4}, \frac{3\pi}{4}, \dots \quad (\text{or } \theta = \pm\frac{\pi}{4}) \\ \text{when } \theta &= \frac{\pi}{4} \text{ sub in (1)} \\ z_1 &= i\tan\left(\frac{\pi}{4}\right) \\ \text{when } \theta &= \frac{3\pi}{4} \text{ sub in (1)} \\ z_2 &= i\tan\frac{3\pi}{4} = i\tan\left(-\frac{\pi}{4}\right) = -i\tan\frac{\pi}{4} \\ \therefore \text{the roots of } (1+z)^2 + (1-z)^2 &= 0 \\ \text{are } z &= \pm i\tan\frac{\pi}{4} \end{aligned}$$



Complex Number Solution

$$z^6 = 1 \rightarrow z = 1 + 0i \text{ is one solution}$$

the 5 other solutions found by rotating by $\frac{\pi}{3}$.
(ie 6 solutions evenly spaced around circle)

$$z^3 = i = e^{i\pi/2}$$

$$z = \left(e^{i\pi/2} \right)^{\frac{1}{3}} \rightarrow z = e^{\frac{\pi i}{6}} \text{ is one solution}$$

the 2 other solutions found by rotating by $\frac{2\pi}{3}$.
(ie 3 solutions evenly spaced around circle)

A



Complex Number Solution

$$\begin{aligned} \text{(i) RHP: } (1+i\tan\theta)^n + (1-i\tan\theta)^n &= \frac{2\cos n\theta}{\cos^n\theta} \quad \cos\theta \neq 0, n \in \mathbb{Z}^+ \\ \text{LHS} &= (1+i\tan\theta)^n + (1-i\tan\theta)^n \\ &= \left(\frac{\cos\theta + i\sin\theta}{\cos\theta} \right)^n + \left(\frac{\cos\theta - i\sin\theta}{\cos\theta} \right)^n \\ &= \frac{(\cos\theta + i\sin\theta)^n}{\cos^n\theta} + \frac{(\cos\theta - i\sin\theta)^n}{\cos^n\theta} \\ &= \frac{\cos^n\theta + i\sin^n\theta + \cos^n\theta - i\sin^n\theta}{\cos^n\theta} \\ &= \frac{2\cos^n\theta}{\cos^n\theta} = \text{RHS} \\ \text{(ii) Let } z &= i\tan\theta, n=2 \\ (1+z)^2 + (1-z)^2 &= 0 \rightarrow \text{Quadratic eqn, 2 roots} \\ \frac{2\cos 2\theta}{\cos^2\theta} &= 0 \\ \cos 2\theta &= 0 \\ 2\theta &= \frac{\pi}{2}, \frac{3\pi}{2} \\ \theta &= \frac{\pi}{4}, \frac{3\pi}{4} \\ z_1 &= i\tan\frac{\pi}{4} \\ z_2 &= i\tan\frac{3\pi}{4} = i\tan\left(-\frac{\pi}{4}\right) = -i\tan\frac{\pi}{4} \\ \therefore z &= \pm i\tan\frac{\pi}{4} \end{aligned}$$



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Complex Number



The complex number $u = 3 - i$ is represented on an Argand diagram by the point A . Points B and C , respectively, represent the complex numbers $\bar{u}$ and $\bar{u} - u$, and O represents the origin.

- Sketch points A , B and C on an Argand diagram. 2
- What type of quadrilateral is $OABC$? 1
- Calculate $\frac{\bar{u}}{u}$, expressing your answer in the form $x + iy$ where x and y are real. 1
- Deduce that $\tan^{-1}\left(\frac{3}{4}\right) = 2 \tan^{-1}\left(\frac{1}{3}\right)$. 2



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Complex Number



Consider the complex number $z = i$.

- Explain why z can be written as $e^{i\frac{\pi}{2}}$. 1
- Hence prove that $i^i \approx 0.208$. 2
- By considering another value of $\arg(i)$ find a value of i^i that is greater than 1. 1



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Complex Number



- Using de Moivre's theorem, show that 3

$$\tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}.$$
- Hence show that $\tan^2\left(\frac{\pi}{5}\right)$ and $\tan^2\left(\frac{2\pi}{5}\right)$ are the roots of the equation $x^2 - 10x + 5 = 0$. 2



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Complex Number



For a complex number z , shade the region of the Argand Plane in which 2

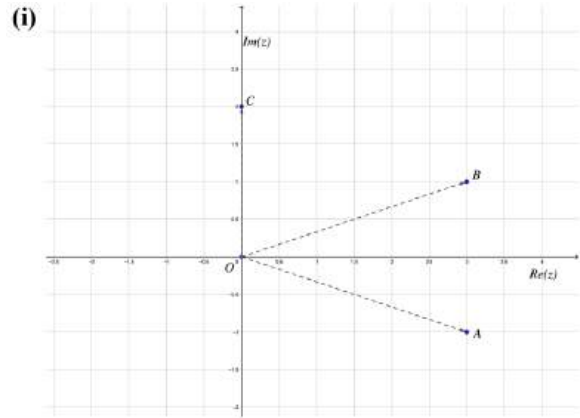
$$|z| < 3 \text{ and } -2 < \operatorname{Im}(z) < 2.$$


$$(i) \quad \arg i = \frac{\pi}{2} \text{ and } |i| = 1, \text{ so } z = 1e^{i\frac{\pi}{2}}$$

$$(ii) \quad i^i = \left(e^{i\frac{\pi}{2}} \right)^i = e^{i^2\frac{\pi}{2}} = e^{-\frac{\pi}{2}} \approx 0.208$$

$$(iii) \quad \arg i = \frac{-3\pi}{2} \rightarrow i = e^{-i\frac{3\pi}{2}}$$

$$i^i = \left(e^{-i\frac{3\pi}{2}} \right)^i = e^{\frac{3\pi}{2}} \approx 111.318$$



(ii) $OABC$ is a parallelogram

$$(iii) \quad \frac{\bar{u}}{u} = \frac{3+i}{3-i} = \frac{3+i}{3-i} \times \frac{3+i}{3+i} = \frac{(3+i)^2}{3^2+1^2} = \frac{9+6i+i^2}{10} = \frac{4}{5} + \frac{3}{5}i$$

$$(iv) \quad \arg\left(\frac{\bar{u}}{u}\right) = \arg \bar{u} - \arg u$$

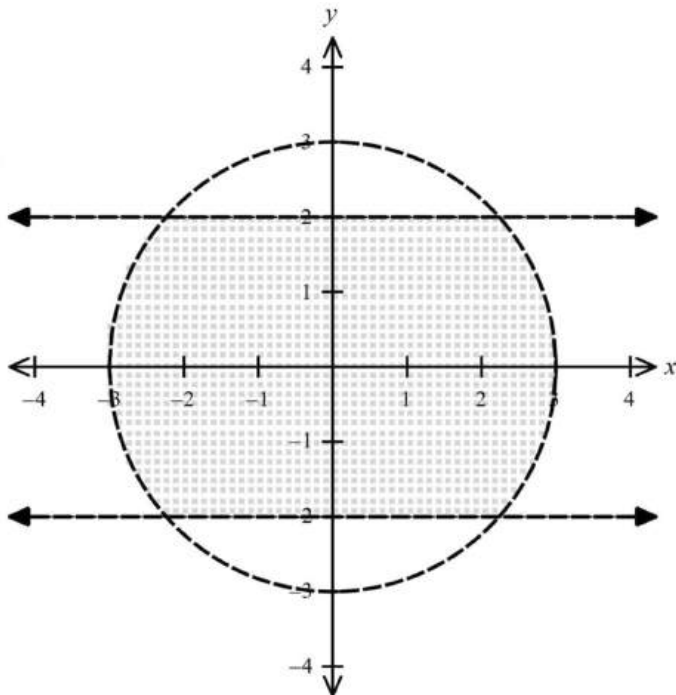
$$\arg\left(\frac{4}{5} + \frac{3}{5}i\right) = \arg(3+i) - \arg(3-i)$$

$$\tan^{-1}\left(\frac{\frac{3}{5}}{\frac{4}{5}}\right) = \tan^{-1}\left(\frac{1}{3}\right) - \tan^{-1}\left(\frac{-1}{3}\right)$$

$$= \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{3}\right)$$

$$\tan^{-1}\left(\frac{3}{4}\right) = 2 \tan^{-1}\left(\frac{1}{3}\right)$$

(or, using
 $2 \arg(\bar{u}) = \arg(\bar{u}^2)$)



(i) By de Moivre's Theorem and Binomial Theorem:

$$\begin{aligned} \cos 5\theta + i \sin 5\theta &= (\cos \theta + i \sin \theta)^5 \\ &= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3 + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5 \\ &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta + i(5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta) \end{aligned}$$

Equating real and Imaginary parts:

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

$$\tan 5\theta = \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

$$= \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

$$= \frac{\cos^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta} \quad \text{* show}$$

$$= \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

(ii) if $\tan 5\theta = 0$

$$\text{then } 5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta = 0$$

$$\tan \theta (5 - 10 \tan^2 \theta + \tan^4 \theta) = 0$$

$$\text{Also, } \tan 5\theta = 0 \rightarrow \theta = \frac{\pi}{5}, \frac{2\pi}{5}$$

$$\text{and, as } \tan \frac{\pi}{5} \neq 0, \tan \frac{2\pi}{5} \neq 0$$

$$\text{then } \theta = \frac{\pi}{5}, \theta = \frac{2\pi}{5} \text{ are solutions of } 5 - 10 \tan^2 \theta + \tan^4 \theta = 0$$

$$\text{Hence } \tan \frac{\pi}{5}, \tan \frac{2\pi}{5} \text{ are solutions of } 5 - 10x^2 + x^4 = 0$$

$$\text{and } \tan^2 \frac{\pi}{5}, \tan^2 \frac{2\pi}{5} \text{ are solutions of } 5 - 10x + x^2 = 0$$



HSC Extension 2 Mathematics

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Complex Number



Prove that for all complex numbers z , $e^z \neq 0$.

3



HSC Extension 2 Mathematics

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Complex Number



Let P be the point which represents the complex number $z = x + iy$ in the complex plane.

- (i) Sketch the curve traced out by P in the complex plane if $|z - 1 - i| = \text{Im}(z + 1 + i)$. **You MUST draw the curve on the axes provided for you on the back of the Multiple Choice sheet.** 3

- (ii) Show that $-(\overline{iz}) = y + ix$. 1

Suppose now that the point Q represents the complex number $-(\overline{iz})$.

- (iii) Sketch the curve traced out by Q on the same diagram as part (i). Show all necessary features. 2



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Complex Number



The polynomial $P(z) = z^4 - 10z^3 + cz^2 + dz + 169$ has two zeroes $\alpha = a + ib$ and $\beta = b + ia$, where $a, b, c, d \in \mathbf{R}$ and $a \neq b$.

- (i) Express the other two zeroes of $P(z)$ in terms of a and b , justifying your answer. 1
- (ii) Find the values of a and b , given that they are both integers. 2



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Complex Number



- (i) Use De Moivre's theorem to show that $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$. 3

- (ii) Consider the equation $x^4 - 4x^2 + 1 = 0$. Use the substitution $x = 2 \cos \theta$ to show that the roots are $2 \cos \frac{\pi}{12}$, $2 \cos \frac{5\pi}{12}$, $2 \cos \frac{7\pi}{12}$, and $2 \cos \frac{11\pi}{12}$. 2

- (iii) Hence show that $\cos^2 \frac{\pi}{12} + \cos^2 \frac{5\pi}{12} + \cos^2 \frac{7\pi}{12} + \cos^2 \frac{11\pi}{12} = 2$. 2



Complex Number Solution

$$(i) |z-1-i| = \text{Im}(z+1+i)$$

$$|(x-1) + (y-1)i| = y+1$$

$$(x-1)^2 + (y-1)^2 = (y+1)^2$$

$$(x-1)^2 + y^2 - 2y + 1 = y^2 + 2y + 1$$

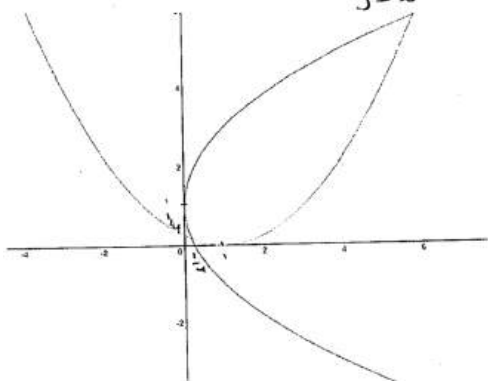
$$y = \frac{1}{4}(x-1)^2$$

$$(ii) -(\overline{iz}) = -(\overline{i(x+iy)})$$

$$= -(ix-y)$$

$$= -(-ix-y) = y+ix$$

(iii) curves are inverse relations. Reflection of each other in line $y=x$



Complex Number Solution

let $z = x+iy$, $x, y \in \mathbb{R}$
and suppose $e^z = 0$

$$\therefore e^{x+iy} = 0$$

$$e^x e^{iy} = 0$$

$$\therefore e^x = 0 \text{ or } e^{iy} = 0$$

$$e^x = 0$$

$$\text{as } x \in \mathbb{R}, e^x \neq 0$$

$$e^{iy} = 0$$

$$|e^{iy}| = 1$$

$$|0| = 0$$

$$\text{as } |e^{iy}| \neq |0|$$

$$\therefore e^{iy} \neq 0$$

$\therefore$ neither e^x or e^{iy} can be 0

$\therefore$ by contradiction, $e^z \neq 0$ for all complex numbers z .

Incorrect methods:

$$e^z = 0$$

$$\ln e^z = \ln 0$$

$$z = \ln 0$$

$\therefore$ no complex number z .

NOTE: $\ln 0$ is only undefined for real numbers.

$$\log(z) = \ln|z| + i \arg(z)$$

this is not part of our course



Complex Number Solution

(i) $(\cos\theta + i\sin\theta)^4 = \cos 4\theta + i\sin 4\theta$ by De Moivre's theorem

$$(\cos\theta + i\sin\theta)^4 = \binom{4}{0}\cos^4\theta + \binom{4}{1}\cos^3\theta(i\sin\theta) + \binom{4}{2}\cos^2\theta(i\sin\theta)^2 + \binom{4}{3}\cos\theta(i\sin\theta)^3 + \binom{4}{4}(i\sin\theta)^4$$

$$= \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta + i(4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta)$$

$$\text{Equating real: } \cos 4\theta = \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$$

$$= \cos^4\theta - 6\cos^2\theta(1-\cos^2\theta) + (1-\cos^2\theta)^2$$

$$= \cos^4\theta - 6\cos^2\theta + 6\cos^4\theta + 1 - 2\cos^2\theta + \cos^4\theta$$

$$= 8\cos^4\theta - 8\cos^2\theta + 1$$

$$(ii) x^4 - 4x^2 + 1 = 0$$

Let $x = 2\cos\theta$:

$$(2\cos\theta)^4 - 4 \times (2\cos\theta)^2 + 1 = 0$$

$$16\cos^4\theta - 16\cos^2\theta + 1 = 0$$

$$8\cos^4\theta - 8\cos^2\theta + \frac{1}{2} = 0$$

$$8\cos^4\theta - 8\cos^2\theta + 1 = \frac{1}{2}$$

$$\cos 4\theta = \frac{1}{2}$$

So the roots of the equation are $x = 2\cos\theta$, where $\cos 4\theta = \frac{1}{2}$

$$\cos 4\theta = \frac{1}{2}$$

$$4\theta = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, 2\pi + \frac{\pi}{3}, 4\pi - \frac{\pi}{3} \text{ (taking further values)}$$

$$= \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3} \text{ gives duplicate results for } \cos\theta$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$$

$$\therefore \text{roots are } 2\cos\frac{\pi}{12}, 2\cos\frac{5\pi}{12}, 2\cos\frac{7\pi}{12}, 2\cos\frac{11\pi}{12}$$

$$(iii) x^2 + \beta^2 + \gamma^2 + \delta^2 = (x+\beta+\gamma+\delta)^2 - 2(\alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta)$$

$$\therefore 4\cos^2\frac{\pi}{12} + 4\cos^2\frac{5\pi}{12} + 4\cos^2\frac{7\pi}{12} + 4\cos^2\frac{11\pi}{12} = 0^2 - 2 \times -4 = 8$$

$$\therefore \cos^2\frac{\pi}{12} + \cos^2\frac{5\pi}{12} + \cos^2\frac{7\pi}{12} + \cos^2\frac{11\pi}{12} = 2$$



Complex Number Solution

(i) The coefficients of $P(z)$ are real, so complex zeroes occur in conjugate pairs.

$\therefore$ the other zeroes are $a-ib$ and $b-ia$.

(ii) Sum of zeroes:

$$(a+ib) + (a-ib) + (b+ia) + (b-ia) = 10$$

$$2a + 2b = 10$$

$$a + b = 5 \quad (1)$$

Product of zeroes:

$$(a+ib)(a-ib)(b+ia)(b-ia) = 169$$

$$(a^2+b^2)(b^2+a^2) = 169$$

$$(a^2+b^2)^2 = 169$$

$$a^2+b^2 = 13, \text{ as } a, b \in \mathbb{R}$$

$$(1)^2: a^2 + 2ab + b^2 = 25$$

$$2ab + 13 = 25$$

$$ab = 6$$

By inspection, as a & b are integers, $a=2, b=3$ (or $b=2, a=3$).



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Complex Number



(i) Write down the roots of the equation $z^7 = 1$.

2

(ii) Hence show that

2

$$z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 = (z^2 - 2z \cos \frac{2\pi}{7} + 1)(z^2 - 2z \cos \frac{4\pi}{7} + 1)(z^2 - 2z \cos \frac{6\pi}{7} + 1).$$

(iii) Deduce that $\sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{8}$.

2



HSC Extension 2 Mathematics

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Complex Number



The complex number z satisfies $|z - 4 - 6i| - |z + 2| = 0$.

What is the minimum value of $|z|$?



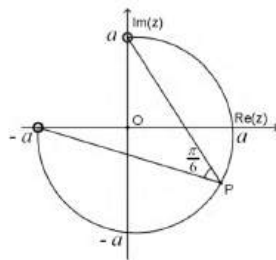
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Complex Number



Which of the equations best represents the given diagram of complex number z , where $a \in \mathbb{Z}$?



NOT TO
SCALE

(A) $\arg\left(\frac{z-a}{z+ai}\right) = \frac{\pi}{6}$

(B) $\arg\left(\frac{z+ai}{z-a}\right) = \frac{\pi}{6}$

(C) $\arg\left(\frac{z+ai}{z-a}\right) = -\frac{\pi}{6}$

(D) $\arg\left(\frac{z+a}{z-ai}\right) = \frac{\pi}{6}$



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Complex Number



(i) By applying de Moivre's theorem and by expanding $(\cos \theta + i \sin \theta)^5$,
show $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$ as a polynomial in $\cos \theta$.

(ii) Solve the equation $\cos 5\theta = -1$ for $0 \leq \theta \leq 2\pi$.

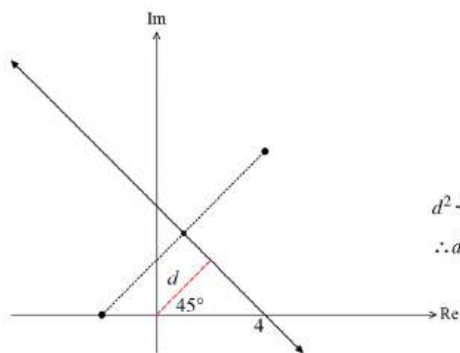
Hence show that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$ and find the value of $\cos \frac{\pi}{5} \cos \frac{3\pi}{5}$.



Complex Number Solution

$$|z - 4 - 6i| = |z + 2| \quad (\text{This is the perpendicular bisector of } 4 + 6i \text{ and } -2)$$

Midpoint is (1, 3), and so the equation is $y = -x + 4$.



$$d^2 + d^2 = 16 \Rightarrow d^2 = 8$$

$$\therefore d = 2\sqrt{2}$$



Complex Number Solution

$$(i) z^7 = 1$$

$$\text{Let } z = cis\theta$$

$$cis7\theta = cis0$$

$$7\theta = 2k\pi$$

$$\theta = \frac{2k\pi}{7}, \quad k=0, \pm 1, \pm 2, \pm 3$$

$\therefore$ the roots are

$$1, cis\frac{2\pi}{7}, cis(-\frac{2\pi}{7}), cis\frac{4\pi}{7}, cis(-\frac{4\pi}{7}), cis\frac{6\pi}{7}, cis(-\frac{6\pi}{7}).$$

$$(ii) z^7 = 1$$

$$z^7 - 1 = 0$$

$$(z-1)(z^6 + z^5 + z^4 + z^3 + z^2 + z + 1) = 0$$

As 1 is a zero of $z^7 - 1 = 0$,

$$\begin{aligned} z^6 + z^5 + z^4 + z^3 + z^2 + z + 1 &= (z - cis\frac{2\pi}{7})(z - cis\frac{4\pi}{7})(z - cis\frac{6\pi}{7})(z - cis\frac{8\pi}{7})(z - cis\frac{10\pi}{7})(z - cis\frac{12\pi}{7}) \\ &= (z^2 - 2z\cos\frac{2\pi}{7} + 1)(z^2 - 2z\cos\frac{4\pi}{7} + 1)(z^2 - 2z\cos\frac{6\pi}{7} + 1) \end{aligned}$$

(iii) Let $z = 1$ in identity from part (ii):

$$1 + 1 + 1 + 1 + 1 + 1 = (2 - 2\cos\frac{2\pi}{7})(2 - 2\cos\frac{4\pi}{7})(2 - 2\cos\frac{6\pi}{7})$$

$$7 = 2^3(1 - \cos\frac{2\pi}{7})(1 - \cos\frac{4\pi}{7})(1 - \cos\frac{6\pi}{7})$$

$$7 = 2^6 \left(\frac{1}{2}(1 - \cos\frac{2\pi}{7})\right) \left(\frac{1}{2}(1 - \cos\frac{4\pi}{7})\right) \left(\frac{1}{2}(1 - \cos\frac{6\pi}{7})\right)$$

$$7 = 2^6 \sin^2\frac{\pi}{7} \times \sin^2\frac{2\pi}{7} \times \sin^2\frac{3\pi}{7}$$

$$\therefore \sin^2\frac{\pi}{7} \sin^2\frac{2\pi}{7} \sin^2\frac{3\pi}{7} = \frac{7}{2^6}$$

$$\sin\frac{\pi}{7} \sin\frac{2\pi}{7} \sin\frac{3\pi}{7} = \frac{\sqrt{7}}{2^3}$$

$$= \frac{\sqrt{7}}{8}$$



Complex Number Solution

$$(i) (\cos\theta + i\sin\theta)^5 = \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta$$

By de Moivre's theorem,

$$\cos 5\theta + i\sin 5\theta = \cos^5\theta + 5i\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10i\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + i\sin^5\theta$$

Equating real terms,

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta = \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)^2$$

$$= \cos^5\theta - 10(\cos^3\theta - \cos^5\theta) + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta)$$

$$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos^5\theta - 10\cos^3\theta + 5\cos\theta$$

$$\therefore \cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta \quad (\text{as required})$$

$$(ii) \cos 5\theta = -1 \text{ for } 0 \leq \theta \leq 2\pi.$$

$$5\theta = 2k\pi + \pi, \quad k=0, 1, 2, 3, 4$$

$$\theta = \frac{2k\pi + \pi}{5} = \frac{(2k+1)\pi}{5}$$

$$\text{When } k=0, \theta = \frac{\pi}{5}, \text{ one root is } \cos\frac{\pi}{5}$$

$$\text{When } k=3, \theta = \frac{7\pi}{5}, \text{ one root is } \cos\frac{7\pi}{5} = \cos\frac{3\pi}{5}$$

$$\text{When } k=1, \theta = \frac{3\pi}{5}, \text{ one root is } \cos\frac{3\pi}{5}$$

$$\text{When } k=4, \theta = \frac{9\pi}{5}, \text{ one root is } \cos\frac{9\pi}{5} = \cos\frac{\pi}{5}$$

$$\text{When } k=2, \theta = \frac{5\pi}{5} = \pi, \text{ one root is } \cos\pi = -1$$

$$\text{Given } \cos 5\theta = -1$$

$$\text{i.e. } 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = -1$$

$$\therefore 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta + 1 = 0$$

$$a=16, b=0, c=-20, d=0, e=5, f=1$$

Sum of single roots:

$$\cos\frac{\pi}{5} + \cos\frac{3\pi}{5} - 1 + \cos\frac{7\pi}{5} + \cos\frac{9\pi}{5} = -\frac{b}{a}$$

$$\cos\frac{\pi}{5} + \cos\frac{3\pi}{5} - 1 + \cos\frac{3\pi}{5} + \cos\frac{\pi}{5} = 0$$

$$2\cos\frac{\pi}{5} + 2\cos\frac{3\pi}{5} = 1$$

$$2\left(\cos\frac{\pi}{5} + \cos\frac{3\pi}{5}\right) = 1$$

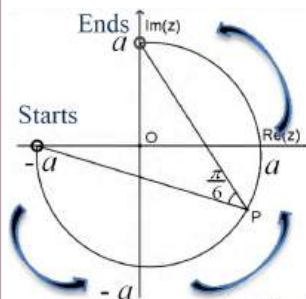
$$\therefore \cos\frac{\pi}{5} + \cos\frac{3\pi}{5} = \frac{1}{2} \quad (\text{as required})$$

Product of 5 roots:

$$\left(\cos\frac{\pi}{5}\right)\left(\cos\frac{3\pi}{5}\right)(-1)\left(\cos\frac{7\pi}{5}\right)\left(\cos\frac{9\pi}{5}\right) = -\frac{f}{a}$$



Complex Number Solution



Anticlockwise direction i.e. positive direction

$$\text{So } \arg(z + ai) - \arg(z - a) = \frac{\pi}{6}$$

$$\arg\left(\frac{z + ai}{z - a}\right) = \frac{\pi}{6} \quad (D)$$



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Complex Number



If ω is the complex cube root of unity, evaluate $(1-3\omega+\omega^2)(1+\omega-8\omega^2)$. 2



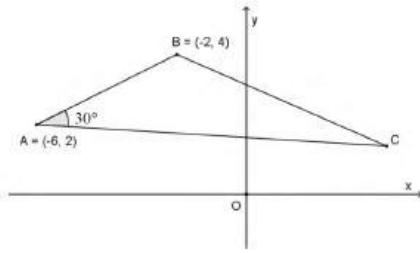
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Complex Number



NOT TO
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$\triangle ABC$ is drawn in the Argand diagram above where $\angle BAC = 30^\circ$, A and B are the points $(-6, 2)$ and $(-2, 4)$ respectively. The length of side AC is twice the length of side AB .

- Show that the complex number that represent vector $\overrightarrow{AB}$ is $4 + 2i$. 1
- Find the complex number that the point C represents. 2
- Find the complex number that the point D represents such that $\triangle ABD$ is an equilateral triangle. 2



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Complex Number



Find the geometrical shape represented by the complex number z if $\omega = \frac{z-2}{z}$, given that ω is purely imaginary. 3



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Complex Number

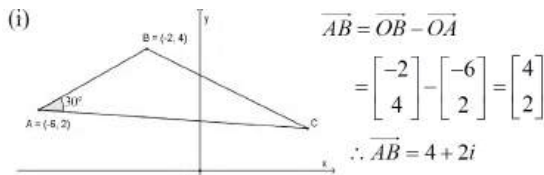


Let $z = e^{i\theta}$ where $z \neq 0$.

- Show that $z^n - \frac{1}{z^n} = 2i \sin n\theta$ for positive integers $n \geq 1$. 2
- Expand $\left(z - \frac{1}{z}\right)^5$ and show that $\sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$. 3
- Find $\int \sin^5 \theta d\theta$. 1



Complex Number Solution



(ii) $\vec{OC} = \vec{OA} + \vec{AC}$ where $|\vec{AC}| = 2|\vec{AB}|$

$$\vec{AC} = 2 \operatorname{cis}\left(-\frac{\pi}{6}\right)(4 + 2i)$$

$$= 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)(4 + 2i) = (\sqrt{3} - i)(4 + 2i)$$

$$= 4\sqrt{3} + 2\sqrt{3}i - 4i + 2 = (4\sqrt{3} + 2) + (2\sqrt{3} - 4)i$$

$$\vec{OC} = \vec{OA} + \vec{AC}$$

$$\vec{OC} = (-6 + 2i) + [(4\sqrt{3} + 2) + (2\sqrt{3} - 4)i]$$

$$\therefore \vec{OC} = (4\sqrt{3} + 2 - 6) + (2\sqrt{3} - 4 + 2)i$$

$$\therefore \vec{OC} = (4\sqrt{3} - 4) + (2\sqrt{3} - 2)i$$

(iii) $\vec{OD} = \vec{OA} + \vec{AD} = \vec{OA} + \vec{AB} \operatorname{cis} \frac{\pi}{3}$

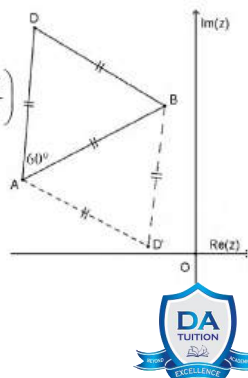
$$= (-6 + 2i) + (4 + 2i)\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$$

$$= (-6 + 2i) + (4 + 2i)\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$

$$= (-6 + 2i) + (2 + i)(1 + \sqrt{3}i)$$

$$= (-6 + 2i) + 2 + 2\sqrt{3}i + i - \sqrt{3}$$

$$\therefore \vec{OD} = (-4 - \sqrt{3}) + (2\sqrt{3} + 3)i$$



Complex Number Solution

Given ω is the complex cube root of unity

i.e. $z^3 = 1$

$$\omega^3 = 1$$

$$\omega^3 - 1 = 0$$

$$(\omega - 1)(1 + \omega + \omega^2) = 0 \quad \text{but } \omega \neq 1$$

$$\therefore 1 + \omega + \omega^2 = 0$$

$$1 + \omega = -\omega^2 \quad \text{or} \quad \omega + \omega^2 = -1 \quad \text{or} \quad 1 + \omega^2 = -\omega$$

$$\therefore (1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2) = ((1 + \omega^2) - 3\omega)((1 + \omega) - 8\omega^2)$$

$$= (-\omega - 3\omega)(-\omega^2 - 8\omega^2)$$

$$= (-4\omega)(-9\omega^2)$$

$$= 36\omega^3 \quad \text{and since } \omega^3 = 1$$

$$= 36$$



Complex Number Solution

(i)

$$z^n = (\cos \theta + i \sin \theta)^n$$

$$= \cos n\theta + i \sin n\theta \quad (\text{de Moivre's theorem})$$

$$\frac{1}{z^n} = (\cos \theta + i \sin \theta)^{-n}$$

$$= \cos(-n)\theta + i \sin(-n)\theta$$

$$= \cos n\theta - i \sin n\theta$$

$$\therefore z^n - \frac{1}{z^n} = (\cos n\theta + i \sin n\theta) - (\cos n\theta - i \sin n\theta)$$

$$= \cos n\theta + i \sin n\theta - \cos n\theta + i \sin n\theta$$

$$\therefore z^n - \frac{1}{z^n} = 2i \sin n\theta \quad (\text{as required})$$

(ii)

$$\left(z - \frac{1}{z}\right)^5 = \binom{5}{0}z^5 - \binom{5}{1}z^4\left(\frac{1}{z}\right) + \binom{5}{2}z^3\left(\frac{1}{z}\right)^2 + \binom{5}{3}z^2\left(\frac{1}{z}\right)^3$$

$$+ \binom{5}{4}z\left(\frac{1}{z}\right)^4 - \binom{5}{5}\left(\frac{1}{z}\right)^5$$

$$= z^5 - 5z^3 + 10z - 10\left(\frac{1}{z}\right) + 5\left(\frac{1}{z}\right)^3 - \left(\frac{1}{z}\right)^5$$

$$\left(z - \frac{1}{z}\right)^5 = \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$$

$$(2i \sin \theta)^5 = (2i \sin 5\theta) - 5(2i \sin 3\theta) + 10(2i \sin \theta)$$

$$32i \sin^5 \theta = 2i(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$$

$$16 \sin^5 \theta = \sin 5\theta - 5 \sin 3\theta + 10 \sin \theta$$

$$\therefore \sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta) \quad (\text{as required})$$

(iii)

$$\therefore \int \sin^5 \theta \, d\theta = \frac{1}{16} \int (\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta) \, d\theta$$

$$= \frac{1}{16} \left(-\frac{\cos 5\theta}{5} + \frac{5 \cos 3\theta}{3} - 10 \cos \theta \right) + C$$

$$= -\frac{\cos 5\theta}{80} + \frac{5 \cos 3\theta}{48} - \frac{5 \cos \theta}{8} + C$$



Complex Number Solution

$$\omega = \frac{z-2}{z} \quad \text{Let } z = x + iy$$

$$\omega = 1 - \frac{2}{z}$$

$$= 1 - \frac{2}{x + iy} \times \frac{x - iy}{x - iy}$$

$$= 1 - \frac{2(x - iy)}{x^2 + y^2}$$

$$= 1 - \frac{2x}{x^2 + y^2} + \frac{2iy}{x^2 + y^2}$$

Since ω is purely imaginary, i.e. $\operatorname{Re}(\omega) = 0$

$$\therefore 1 - \frac{2x}{x^2 + y^2} = 0$$

$$\frac{x^2 + y^2 - 2x}{x^2 + y^2} = 0$$

$$x^2 + y^2 - 2x = 0$$

$$x^2 - 2x + y^2 = 0$$

$$x^2 - 2x + 1 + y^2 = 0 + 1$$

$$(x - 1)^2 + y^2 = 1$$

$$\therefore |z - 1| = 1$$

Hence the locus of z is a circle of radius 1 and centre $(1, 0)$

